SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: CSA1402 **COURSE NAME: Compiler Design**

COURSE OUTCOME COVERED

CO2: *Construct top-down and bottom up parsers using CFG for parsing conflicts and error recovery for efficient syntax tree generation (BL3)*

Assessment Tool Weightage: 20%

QUESTIONS: 20 **Total Marks:100**

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Understanding of Concepts	25%				
Explanation	25%				
Application	20%				
Organization	15%				
Accuracy	10%				
Clarity	5%				

Faculty Incharge

Annexure A

SHORT ANSWER TEST

1. Construct language for the following Regular Expression:2

   (a|b)*a(a|b)|(a|b)

2. a) Illustrate the predictive parser for the following grammar. 4

   S → (L) | a
   L → L, S | S

3. Analyse, is it possible, by modifying the grammar in any way to construct predictive parser for the language of
   S → SS+ | SS* | a
   String: "$aa+a$" 4

4. Eliminate left recursion from the grammar $S \rightarrow Sa|Sb|c$ 2
5. The following grammar contains common prefixes, making it unsuitable for predictive parsing.

   A→ abcD | abcE | f

   Apply left factoring to remove the common prefixes and rewrite the grammar.5

6. Consider the following statement:10
   $$Degree - F = Degree - C * 1.8 + 32$$
   Show the representation of this statement after each phase of the compiler and explain.
7. Consider the following C code fragment:4
   ```
   sum = 0;

   for (i = 1; i < 5; i++)
       sum = sum + i;
   ```
   Show the representation of the above code after each phase of the compiler.2

8. Consider the following grammar.
   $$S \rightarrow aABb,\ A \rightarrow c|\varepsilon,\ B \rightarrow d|\varepsilon$$
   Calculate FIRST and FOLLOW, Obtain the Predictive Parsing Table.
   not. Show the parser moves for the input string **acdb.**
9. Write a simple Lex program to recognize and count identifiers in a given input.2
10. Write the rule to Eliminate Left Recursion and Left Factoring.2
11. What are the actions performed by a shift-reduce parser? Explain each action with a suitable example.3
12. Consider the following grammar:
    $$S \rightarrow pBB \rightarrow qB|r$$
    Find the leftmost derivation and construct the parse tree for the input string `pqqr`.3
13. Consider Grammar: E-> E+T|T T-> T*F|F F-> (E)|id
    Eliminate left recursion and find FIRST and FOLLOW 5
14. **For the following grammar, determine whether the string `abcc` is accepted using shift-reduce parsing. Show all shift and reduce operations. 10**

$$S \to AB A \to aA | aB \to cB | c$$

15. For the following grammar, verify whether the string `abbc` is valid using shift-reduce parsing.10

$$S \to aAB A \to bA | bB \to c$$

Show the complete sequence of stack, input, and parser actions.

16. Consider the following grammar:10

$$S^{'} \to SS \to aAB A \to bA | cB \to d$$

- Construct the canonical collection of LR(0) items.
- Construct the SLR(1) parsing table.

17. Consider the following grammar.

$$S \to iEtSS^{'} | a,\ S^{'} \to eS | \varepsilon,\ E \to b$$

Obtain the Predictive Parsing Table. 5

18. Do left factoring in the following grammar: 3

A-> aAB|aA|a

B-> bB|b

19. Construct the LL(1) Parsing Table for the following grammars.7

$$S \to ABC A \to a | \varepsilon B \to r | \varepsilon C \to b | \varepsilon$$

20. Construct the LL(1) Parsing Table for the following grammars.7

$$S \to 1S\# | qABC A \to a | bbD B \to a | \varepsilon C \to b | \varepsilon D \to c | \varepsilon$$

RUBRICS FOR CALCULATION

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers

Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand